

More than 67.3195% of the zeros of the Riemann zeta function are
simple and lie on the critical line

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Abstract

Let $N(T, 2T)$ count the zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in that interval. We prove

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.673195198901\dots > \frac{673195}{10^6}.$$

The proof combines the analytic framework of Claude [1], the stability refinement of the rank-trace argument introduced in [2], and the re-optimized window, position-weighted seven-point inequality, and sharp block profile of [3]. The single new mathematical ingredient is a strengthened certified constant for the weighted seven-point inequality of [3]: we certify $F \geq 509/100000$ where $1/200$ was certified before, and re-optimize the block length. The finite claims are certified by rigorous interval computations, with the principal certificate repeated at two grid resolutions; the deduction is reproduced in full.

1 Statement and imported inputs

Throughout, write $N := N(T, 2T)$ for the number of zeros of ζ with ordinates in $(T, 2T]$, counted with multiplicity, and $S := N_0^s(T, 2T)$ for the simple zeros on the critical line in that range.

Theorem 1.1.

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{250 H_{\text{cert}} - \eta \frac{3}{1150} \cdot 249}{250 - R} = 0.6731951989015205\dots > \frac{673195}{10^6},$$

where $H_{\text{cert}} = 672457/10^6$, $A = 31049/25000$, $R = 2\sqrt{A} - 1$, and $\eta = R/A$.

The proof template is that of [3], which in turn builds on [2] and [1]. We isolate below the two external analytic consequences used in the finite deduction. The first is elementary from the Riemann–von Mangoldt formula; the second combines the arbitrary- window trace asymptotics of [1, §7.1] with the stability rank–trace refinement of [2].

Imported Input 1.2 (Normalized ordinates and total span). *Put $\ell_T = \log(T/(2\pi))$ and, for a zero of ordinate γ , put*

$$x_\gamma = \frac{\ell_T(\gamma - T)}{2\pi}.$$

After the deletion of $o(N)$ exceptional zeros, the retained simple zeros have normalized ordinates $x_1 < \dots < x_{S^\circ}$ with $S^\circ = S - o(N)$ and

$$x_{S^\circ} - x_1 \leq \frac{T\ell_T}{2\pi} = N + o(N). \quad (1)$$

Here the second relation is asymptotic. In the present notation the Riemann–von Mangoldt formula gives

$$N(T, 2T) = \frac{T}{2\pi}(\ell_T + 2\log 2 - 1) + O(\log T) = \frac{T\ell_T}{2\pi} + O(T) = \frac{T\ell_T}{2\pi} + o(N).$$

Imported Input 1.3 (Stability counting inequality). *Let $v : [-\frac{1}{2}, \frac{1}{2}] \rightarrow [\frac{3}{4}, 1]$ be an even C^2 function that is nonincreasing on $[0, \frac{1}{2}]$. At the endpoint $\lambda = 1$, put $\mathcal{L} = \ell_T$, fix a ramp width $w_0 \geq 1$ independently of T , and use $\sqrt{v(u/\mathcal{L})}$ in the test family of [1, §2], multiplied by its standard even boundary ramp of support $[-\mathcal{L}/2, \mathcal{L}/2]$. The required condition $8w_0 \leq \mathcal{L}$ then holds for all sufficiently large T . This is an admissible window provided the variation and derivative conditions recorded in [1, §7.1] hold; the endpoint $\lambda = 1$ is explicitly allowed by [1, Remark 6.1]. Write*

$$K_v(x) = \int_{-1/2}^{1/2} v(t) \cos(2\pi xt) dt, \quad k_v = \frac{K_v}{K_v(0)}, \quad H(v) = 2 - \frac{1}{c_1(v)},$$

where $c_1(v) = (\int v)^2 / (\int v^2 + \iint |s - s'| v(s) v(s') ds ds')$. Then

$$S \geq H(v) N + \text{tr } \Psi(M) - o(N), \quad (2)$$

where M is the Gram matrix of the retained simple-zero atoms, $\Psi(t) = (t - 1)^2$ for $0 \leq t \leq 2$ and $2t - 3$ for $t \geq 2$, and the entries of M satisfy $M_{ij} = k_v(x_i - x_j) + o(1)$ uniformly over all retained pairs in the dyadic interval whose normalized ordinate differences lie in any fixed compact set, the exceptional $o(N)$ zeros being deleted.

We use one retained set for both imported inputs, enlarging the exceptional set to the union of the two $o(N)$ sets if necessary; it still has size $o(N)$. This further deletion replaces the original Gram matrix by a principal compression. Pinching into retained and discarded principal blocks, followed by $\Psi \geq 0$, gives

$$\text{tr } \Psi(M_{\text{original}}) \geq \text{tr } \Psi(M_{\text{retained}}) + \text{tr } \Psi(M_{\text{discarded}}) \geq \text{tr } \Psi(M_{\text{retained}}),$$

so the stability inequality remains valid with the smaller retained matrix. Further deletion preserves the order of the ordinates and can only decrease their outer span; thus Input 1.2 also remains valid for this common retained set, whose consecutive gaps are used below.

Input 1.3 is the stability form of the rank–trace argument: the analytic content (explicit formula, Gabor trace asymptotics for admissible windows, tail estimates, and Gram-entry asymptotics) is from [1]; retention of the defect $\text{tr } \Psi(M)$ is the contribution of [2], with the combined interface also used in [3]. The deduction below uses (2) only through the block/pinching mechanism of Section 5.

2 The window

We use the window of [3] unchanged:

$$v(s) = \sum_{j=0}^6 c_j \cos(\omega_j s), \quad \omega = (\sqrt{2}, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi, 12\pi), \quad (3)$$

$$c = (10^9, 3322500, -7609135, 1190194, -731476, -1680572, 1141360)/10^9.$$

Proposition 2.1 (Certified window bounds). *On $[-\frac{1}{2}, \frac{1}{2}]$ one has $\frac{3}{4} \leq v \leq 1$ (certified global bounds: $v \geq 0.750213\dots$ and $v \leq 0.995632\dots$), and on $[0, \frac{1}{2}]$ one has $v' \leq 0$. Moreover,*

$$H(v) = 0.67245704141454\dots \geq H_{\text{cert}} := \frac{672457}{10^6}.$$

Verification. Arb interval arithmetic over the exact defining data. The range enclosure uses 8192 cells. For monotonicity, the verifier proves $v'' < 0$ on the contiguous union of the four cells adjacent to zero; since the cosine sum is even, $v'(0) = 0$, so $v' < 0$ away from the origin on that union. It proves $v' < 0$ directly on every remaining cell. The enclosures are reproduced by the **fast** module of the verifier (see Section 6). The bounds for v and $H(v)$ reproduce the corresponding certificate of [3]; the derivative check makes explicit the admissibility condition used here. $\square$

We now verify the applicability clause in Input 1.3. Let $\rho \in C^3(\mathbb{R})$ be the fixed nondecreasing function of [1, §2.2], equal to zero on $(-\infty, 0]$ and to one on $[1, \infty)$. Define the smooth even boundary ramp by

$$\rho_{\mathcal{L}, w_0}(u) = \rho\left(\frac{\mathcal{L}/2 - |u|}{w_0}\right)$$

and put

$$\varphi_{\mathcal{L}, w_0}(u) = \begin{cases} \sqrt{v(u/\mathcal{L})} \rho_{\mathcal{L}, w_0}(u), & |u| \leq \mathcal{L}/2, \\ 0, & |u| > \mathcal{L}/2. \end{cases}$$

Since $v \geq 3/4$, its square root is smooth. Proposition 2.1 and the defining properties of the ramp show that both $\varphi_{\mathcal{L}, w_0}$ and $\varphi_{\mathcal{L}, w_0}^2$ are even, nonnegative, compactly supported, C^2 , bounded by one, and nonincreasing for $u \geq 0$. Their total variations are therefore $2\varphi_{\mathcal{L}, w_0}(0) \leq 2$ and $2\varphi_{\mathcal{L}, w_0}^2(0) \leq 2$. On the fixed-width boundary strips, the product rule and the ramp estimates give $\|\varphi_{\mathcal{L}, w_0}''\|_1 + \|(\varphi_{\mathcal{L}, w_0}^2)''\|_1 = O(1/w_0)$; in the interior, scaling the fixed smooth functions $\sqrt{v}$ and v contributes only $O(1/\mathcal{L})$. Thus, for fixed w_0 and all sufficiently large $\mathcal{L}$, this family satisfies the support, variation, and second-derivative hypotheses of [1, §7.1]. This closes the arbitrary-window applicability step rather than treating it as an additional numerical assumption. Finally, the boundary strips have total length $O(w_0)$, while the full support has length $\mathcal{L}$. Their contribution to the normalized moments in (7.1) of [1] is therefore $O(w_0/\mathcal{L}) = o(1)$; likewise the normalized autocorrelation kernel converges, uniformly for bounded normalized shifts, to k_v by the L^1 bound on the ramp discrepancy. Thus the limiting constant and kernel in Input 1.3 are precisely $H(v)$ and k_v displayed above.

3 The sharp block profile

Lemma 3.1. *Let $G \succeq 0$ be Hermitian with $E := 2\sum_{i < j} |G_{ij}|^2$. Then*

$$\text{tr } \Psi(G) \geq h(E), \quad h(E) = \begin{cases} E, & 0 \leq E \leq 1, \\ 2\sqrt{E} - 1, & E \geq 1, \end{cases}$$

and h is sharp.

Proof. For $\lambda \geq 0$ write $x = (\lambda - 1)^2$. Since $\Psi(\lambda) = (\lambda - 1)^2 - ((\lambda - 2)_+)^2$ and, for $\lambda \geq 1$, $(\lambda - 2)_+ = (\sqrt{x} - 1)_+$ while for $0 \leq \lambda < 1$ both $(\lambda - 2)_+ = 0$ and $x \leq 1$, one has the exact identity $\Psi(\lambda) = g((\lambda - 1)^2)$ with

$$g(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2\sqrt{x} - 1, & x \geq 1. \end{cases}$$

The function g is nonnegative, increasing, concave, and $g(0) = 0$; hence g is subadditive, so for the eigenvalues λ_i of G ,

$$\mathrm{tr} \Psi(G) = \sum_i g((\lambda_i - 1)^2) \geq g\left(\sum_i (\lambda_i - 1)^2\right) = g(\|G - I\|_F^2) \geq g(E),$$

using that $\|G - I\|_F^2 = \sum_i (\lambda_i - 1)^2$ and that the off-diagonal part alone contributes E . For sharpness, put $t_n = \sqrt{E(n-1)/n}$, $a_n = 1 + t_n$, and $b_n = 1 - t_n/(n-1)$, taking n large enough that $b_n \geq 0$. With $u_n = n^{-1/2}(1, \dots, 1)$, set $G_n = b_n I + (a_n - b_n)u_n u_n^*$. This matrix is positive semidefinite with constant diagonal one and off-diagonal energy exactly E . Its eigenvalues are $a_n, b_n, \dots, b_n$, and therefore

$$\mathrm{tr} \Psi(G_n) = \Psi(1 + t_n) + (n-1)\Psi(1 - t_n/(n-1)) \longrightarrow h(E).$$

Thus no larger lower profile is possible even among correlation matrices. $\square$

Corollary 3.2 (Chord bound). *For $0 \leq E \leq A$ one has $h(E) \geq \frac{h(A)}{A} E$; and h is nondecreasing.*

Proof. The first assertion is the chord inequality for the concave function h between $(0,0)$ and $(A, h(A))$; monotonicity is immediate from its displayed formula. $\square$

4 The certified weighted seven-point inequality

Let $w = k_v^2$ for the window (3), let $g_1, \dots, g_6 \geq 0$ be consecutive normalized gaps with partial sums $y_j = g_1 + \dots + g_j$ ($y_0 = 0$), and define

$$F(g_1, \dots, g_6) = \frac{1}{2300} \sum_{i=1}^6 g_i + \sum_{0 \leq i < j \leq 6} a_{ij} w(y_j - y_i), \quad (4)$$

with the reflection-symmetric exact rational weights a_{ij} of Table 1. Every span capacity is exactly $\sum_i a_{i,i+r} = 2$ ($r = 1, \dots, 6$).

$10^6 a_{ij}$	$j = 1$	2	3	4	5	6
$i = 0$	239252	528172	965879	1000000	1000000	2000000
1		381335	465776	34121	0	1000000
2			379413	12104	34121	1000000
3				379413	465776	965879
4					381335	528172
5						239252

Table 1: The pair weights of [3], used unchanged.

Proposition 4.1 (Certified local inequality). *For all $g_1, \dots, g_6 \geq 0$,*

$$F(g_1, \dots, g_6) \geq \varepsilon_0 := \frac{509}{100000} = 0.00509. \quad (5)$$

Verification. The pressure term first reduces the unbounded orthant to a compact simplex: if $\sum_i g_i \geq \varepsilon_0/p = 11707/1000$, then the pressure alone proves (5). On the complementary simplex, use exhaustive interval subdivision with Arb enclosures of w on a grid of mesh $1/4000$, exactly the

decision procedure of [3] with the target raised from $1/200$ to $509/100000$: one-body pruning, per-box range-minimum lower bounds over all 21 pair separations, a certified convex tangent bound, and bisection of the widest coordinate; the run fails loudly at any unresolved terminal cell. The recorded run has 1,739,356 nodes and maximum depth 57; the weaker target $127/25000$ was additionally certified in separate regression runs. See Section 6 for the recorded certificates. (The observed minimum of F is ≈ 0.005091029 , at a lattice-like configuration.) $\square$

5 Deduction of the main theorem

Set

$$q = 6, \quad m = 250, \quad p = \frac{1}{2300}, \quad B_p = 6p = \frac{3}{1150},$$

$$A = \varepsilon_0 (m - q) = \frac{509}{100000} \cdot 244 = \frac{31049}{25000} = 1.24196, \quad R = h(A) = 2\sqrt{A} - 1, \quad \eta = \frac{R}{A}.$$

Numerically $R = 1.22886518210501\dots$, $\eta = 0.98945632879079\dots$

5.1 Windows to blocks

Let B be a block of m consecutive retained simple zeros with normalized ordinates $y_1 < \dots < y_m$, Gram matrix G_B , and $E_B = 2 \sum_{i < j} |(G_B)_{ij}|^2$. Summing (5) over the $m - q$ consecutive seven-point windows of B : a pair spanning r gaps lies in at most $7 - r$ windows and its weight totals at most the span capacity 2; each gap lies in at most 6 windows, contributing at most $6p \sum g_i = B_p \text{span}(B)$. By the nonnegativity of the weights, this gives the purely kernel-level bound

$$2 \sum_{i < j} w(y_j - y_i) + B_p \text{span}(B) \geq A. \quad (6)$$

5.2 Blocks to defect

We claim

$$\text{tr } \Psi(G_B) + \eta B_p \text{span}(B) \geq R - o(1). \quad (7)$$

Put

$$D := A/B_p = 714127/1500.$$

If $\text{span}(B) \geq D$, (7) is immediate: $\text{tr } \Psi(G_B) \geq 0$ and $\eta B_p \text{span}(B) \geq \eta A = R$. We may therefore suppose that $\text{span}(B) < D$. Every pair separation in B then lies in the fixed compact interval $[0, D]$. The Gram-entry asymptotic of Input 1.3 is uniform there. More explicitly, its error is bounded by some $\delta_T(D) \rightarrow 0$ for every pair in every such block. Since m is fixed, $|k_v| \leq 1$ because $v \geq 0$, and the same asymptotic gives $|(G_B)_{ij}| \leq 2$ for all sufficiently large T . Consequently

$$2 \sum_{i < j} w(y_j - y_i) \leq E_B + O(m^2 \delta_T(D)) = E_B + o(1).$$

Together with (6), this gives

$$E_B + B_p \text{span}(B) \geq A - o(1). \quad (8)$$

If $E_B \geq A$, then Lemma 3.1 and monotonicity give $\text{tr } \Psi(G_B) \geq h(E_B) \geq h(A) = R$. If $E_B < A$, the chord bound (Corollary 3.2) gives $\text{tr } \Psi(G_B) \geq \eta E_B$, and (8) yields $\eta E_B + \eta B_p \text{span}(B) \geq \eta A - o(1) = R - o(1)$. Because m and D are fixed, the $o(1)$ here is uniform over the small-span blocks; the large-span case has no asymptotic error.

5.3 Pinching and averaging

Exactly as in [2, §5] (and [3, §4]): for each of the m offsets, partition the retained simple zeros into consecutive full blocks of size m , leaving at most $2(m-1)$ endpoint points. If $\mathcal{P}_r M$ denotes the resulting block-diagonal pinching (including the remaining endpoint block), then pinching is an average of unitary conjugations. Convexity, unitary invariance, and $\Psi \geq 0$ therefore give the precise inequality

$$\mathrm{tr} \Psi(M) \geq \mathrm{tr} \Psi(\mathcal{P}_r M) \geq \sum_{B \text{ full at offset } r} \mathrm{tr} \Psi(G_B).$$

The average number of full blocks over the offsets is $S^\circ/m + O(1) = S/m + o(N)$. There are $O(N)$ blocks because $S \leq N$, so the uniform per-block $o(1)$ above sums to $o(N)$. Each interior gap lies in a block span for at most $m-1$ offsets, and Input 1.2 bounds their total length by $N + o(N)$. Summing (7) over full blocks and averaging over the offsets therefore gives

$$\mathrm{tr} \Psi(M) \geq \frac{R}{m} S - \eta B_p \frac{m-1}{m} N - o(N). \quad (9)$$

5.4 Conclusion

Insert (9) and Proposition 2.1 into (2):

$$S \geq H_{\mathrm{cert}} N + \frac{R}{m} S - \eta B_p \frac{m-1}{m} N - o(N),$$

hence

$$\liminf_{T \rightarrow \infty} \frac{S}{N} \geq \frac{m H_{\mathrm{cert}} - \eta B_p (m-1)}{m - R} = \frac{250 \cdot \frac{672457}{10^6} - \eta \cdot \frac{3}{1150} \cdot 249}{250 - R} = 0.6731951989015205755 \dots$$

The final arithmetic (including the enclosure of R and η and the comparison with $673195/10^6$) is itself certified in Arb by the verifier's **fast** module. This proves Theorem 1.1. $\square$

If $\mathcal{N}(X)$ counts zeros up to height X with multiplicity and $\mathcal{S}(X)$ counts the simple critical-line zeros up to height X , the same constant also satisfies

$$\liminf_{X \rightarrow \infty} \frac{\mathcal{S}(X)}{\mathcal{N}(X)} \geq 0.6731951989015205 \dots$$

Indeed, decompose $(0, X]$ into dyadic intervals $(X/2^{j+1}, X/2^j]$. For any fixed error tolerance, Theorem 1.1 applies to every interval whose lower endpoint is sufficiently large; summing their inequalities shows that the omitted fixed bottom interval is negligible as $X \rightarrow \infty$, and then the error tolerance may tend to zero. This is the standard dyadic-summation passage from the displayed theorem to the density formulation in the title.

5.5 Sanity gates

With $(\varepsilon_0, p, m) = (1/200, 1/2300, 257)$ the same formula gives $0.673137630699 \dots$, the bound of [3]; with the unit-cap profile ($\eta = 1, R = A \leq 1$) and the parameters of [2] ($\varepsilon_0 = 19/5000, p = 1/3000, m = 269$, Montgomery–Taylor window) it gives $0.673008527927 \dots$. These are regression checks on the deduction; they are not additional inputs to the present bound.

6 Reproducibility

All finite claims (Propositions 2.1 and 4.1 and the final arithmetic) are certified by rigorous interval computation. Arb [4] encloses the transcendental evaluations. The range- minimum path stores outward-rounded binary64 lower bounds derived from Arb intervals and combines nonnegative terms with directed rounding; the tangent path separately uses Arb enclosures for signed derivatives, gradients, and the LDL convexity test. Source, recorded certificates, and instructions are at

<https://github.com/npip99/zeta-zeros>

These computations do not certify the external analytic statements isolated as Inputs 1.2 and 1.3; those remain dependencies on the cited sources, as is explicit in Section 1. The verifier is that of [3], vendored under MIT with the design constants and an additional monotonicity check changed. As a resolution cross-check, the same decision procedure certified Proposition 4.1 twice: the headline target 509/100000 at grid 1/4000 (1,739,356 nodes, depth 57) and again at grid 1/8000 (1,736,620 nodes, depth 57), the intermediate target 127/25000 also certified in two separate runs (including a 1,544,746-node run during the design exchange). These repeated runs share an implementation and are not claimed to be independent formal proofs. The grid parameter controls the one-dimensional kernel enclosure tables, not the initial six-dimensional box cover, so a similar subdivision-tree size at the two resolutions is expected.

Credit

This note is a minimal increment on [3] (whose window, weights, profile, and base verifier are reused), which in turn extends [2] over the framework of [1]. The improvement was found and cross-verified in a Claude–GPT-5.6 collaboration on August 11, 2026.

References

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